

Math 150A Section 6

Fall 2020

Final Exam Practice

December 16, 2020

Time Limit: 1 hour and 50 Minutes

Name (Print): _____

Student ID: _____

This exam contains 9 pages (including this cover page) and 8 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

You may *not* use your books or notes on this exam. However, you may use a single, handwritten, one-sided notesheet and a *basic* calculator.

You are required to show your work on each problem on this exam, unless otherwise stated. The following rules apply:

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit.** A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit. This especially applies to limit calculations.
- If you need more space, use the back of the pages; clearly indicate when you have done this.
- **Box Your Answer** where appropriate, in order to clearly indicate what you consider the answer to the question to be.
- Unless otherwise stated, all angles are in **radians**

Problem	Points	Score
1	20	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
8	10	
Total:	90	

Do not write in the table to the right.

1. (20 points) **TRUE or FALSE!** If the statement is true, write TRUE (all caps, all spelled out). Otherwise write FALSE (all caps, all spelled out).

(a) $f(x) = |x|$ has a critical point

(b) $\cos(x)$ is concave down at $x = 0$

(c) if $f'(2)$ exists, then $f(x)$ is continuous at $x = 2$

(d) if $f'(1/2) = 2$ then the derivative of $f(\sin(x))$ at $x = \pi/6$ is $\sqrt{3}$

(e) the functions $f(x) = x$ and $g(x) = x^2/x$ are the same

(f) $\int_0^1 f(x)^2 dx = \left(\int_0^1 f(x) dx \right)^2$

(g) $\frac{d}{dx} \sin(2x) = \cos(2x)$

(h) if $f'(x) = g'(x)$ then $f(x) = g(x)$

(i) the function $x^{100} - 9x^2 + 1$ has a root in the interval $[0, 2]$

(j) a function can never cross its horizontal asymptote

2. (10 points) Calculate the following. Carefully show your work.

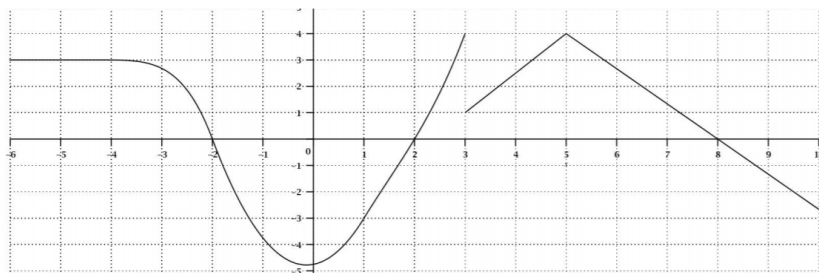
(a) $\lim_{x \rightarrow 2} \frac{\sqrt{x+6} - \sqrt{3x}}{x-3}$

(b) $\lim_{x \rightarrow 1} \frac{\frac{x+3}{x} - \frac{4}{x}}{x-1}$

(c) $f'(x)$ for $f(x) = e^x \sin(x)$

(d) $g'(x)$ for $g(x) = \ln\left(\frac{1+x}{1-x}\right)$

3. (10 points) The following is a graph of the **derivative** $f'(x)$ of a function $f(x)$.



- (a) List all intervals where $f(x)$ is decreasing
- (b) List all intervals where $f(x)$ is concave up
- (c) List all critical points of $f(x)$ and classify them as local max, local min or neither
- (d) What is $\lim_{x \rightarrow 8} \frac{f'(x)}{x-8}$?
- (e) If $f(0) = 4$, is $f(8)$ positive or negative?

4. (10 points)

1. Find all critical points of the function $f(x) = xe^{-x}$

2. If $f(2) = 0.75$ and $f'(2) = -0.25$, use the linearization of $f(x)$ at $x = 2$ to estimate $f(1.9)$.

5. (10 points) Consider functions $f(x)$, $g(x)$, and $h(x)$ whose values satisfy

x	1	2	3	4	5	6	7
$f(x)$	1	1	2	3	5	8	13
$f'(x)$	0	1	3	2	7	9	4

x	1	2	3	4	5	6	7
$g(x)$	8	6	7	5	3	0	9
$g'(x)$	0	1	0	1	2	3	5

x	1	2	3	4	5	6	7
$h(x)$	2	3	5	7	1	13	17
$h'(x)$	4	3	2	1	2	3	4

Evaluate the following expressions.

(a) $F'(5)$ for $F(x) = f(x) - 2g(x) + 3h(x)$

(b) $F'(3)$ for $F(x) = xg(x)$

(c) $F'(1)$ for $F(x) = f(x)g(x)h(x)$

(d) $F'(2)$ for $F(x) = x^{f(x)}$

(e) $F'(4)$ for $F(x) = \frac{g(x)}{x^2 f(x)}$

6. (10 points) Consider an isosceles triangle with base width of 40 and height of 60. What is the largest possible area of a rectangle inscribed in this triangle?

7. (10 points) Find the area of the region bounded by the graphs $y = x^2 - x - 1$ and $y = 1 - x^2$ in the x, y -plane.

8. (10 points) Compute each of the following integrals. Show all work. Lack of work results in no credit!!

(a) $\int_1^8 \frac{3+\sqrt{t}}{t} dt$

(b) $\int \frac{2x}{\sqrt{x^2+1}} dx$

(c) $\int (2z+1) \cos(z^2+z) dz$

(d) $\int_0^3 (\sqrt{9-x^2} + x + 1) dx$